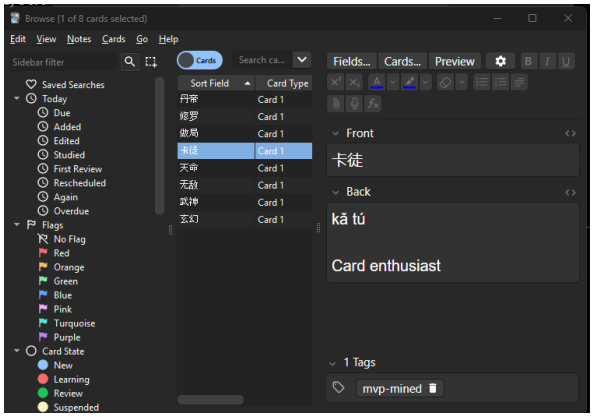
Yomitan at Home

K-Pop Demon Hunters McDonalds Meal

Ok so some IRL friends may have thought that I would talk about K-Pop Demon Hunters since I literally asked about the movie AND had their McDonalds collab meal AND will go to a concert where some of their songs might be played.

However I spent my weekend watching food reels, Valorant games, and League games so here's a low effort blog post!!!



Mandarin Yomitan + Associated Anki Deck

So to lay the groundwork let me explain the #lore of how this project came up. Over the past week or so I've been chatting with some friends about learning new languages. Eventually one decided to post their updates on Discord and overtime I felt like I should try out some of their techniques since we both want to eventually read Japanese source material (iykyk). After a bit of one handed chatting I decided to try out spaced repetition via Anki (a flashcard app) and a 3rd party plug-in called Yomitan to introduce new words into my flashcard deck.

Before I go on to the next part of the story I'll just quickly explain some things below:

- **Spaced repetition** is the technique where you review concepts at a gradually longer time period in an effort to first get a topic into short term memory and then long term memory. For example a word might be reviewed every day for a bit, then every week, and eventually every month.
- **Anki** is a pretty goated flashcard application that's free on desktop and Android with the funds coming in from generous patrons and the iOS app. In addition to allowing spaced repetition it also allows many addons/plugins to be utilized
- **Yomitan** is one of those 3rd party plugins that aims to streamline the process of adding new words into your Anki decks. Specifically it introduces the idea of a **mining deck** which solely comprises of words you see online and allows you to fill in this deck with a web extension. By right clicking a selection you can effectively generate a flashcard on the fly with all the additional info generated by the Yomitan extension. (If you're interested a setup doc is here)

While most of this works completely fine for me it turns out Yomitan is designed exclusively for Japanese and not Chinese (Simplified) which was the language I had in mind. While I'm sure there are mods or versions of Yomitan that support Chinese I decided it would be more fun to build my own since Yomitan also requires some additional finangling that I decided wasn't worth my time.

Fast forward to April 12th at 12:28AM where my initial README commit was pushed to main and I began my very quick development of this project. At a high level I aimed to recreate the user experience of the web extension Yomitan (right click selection and pick the Yomitan specific option) and from there generate a flashcard in Anki containing the original Chinese characters selected at the front and the pinyin + translation on the back.

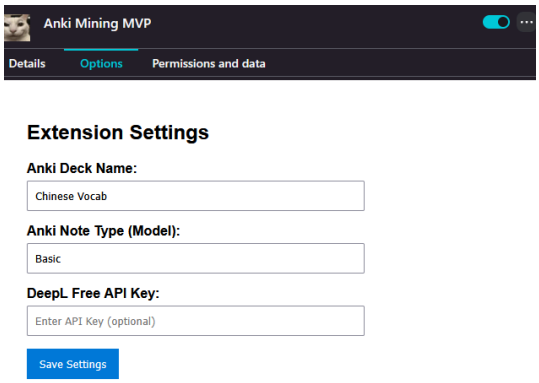
Now it's 1:34AM and I've gotten the high level design down and a functional extension running on Firefox. The pinyin component was relatively trivial since a popular node package already existed called pinyin-pro. Translating the words took a little more effort since I wanted to have an offline solution that the extension would check first before calling any APIs. This was partially to conserve API credits but it was mostly because I figured most of the words I'd add would be considered basic. Eventually I had the offline implementation completed which checks the CC-CEDICT json object first and then if there were no matches the code then calls an API. (Fun fact I actually had to write a quick Python script to generate one big JSON blob since most repos online split up the dictionary into many folders).

After waking up and eating lunch I went back into the flow state and eventually completed the entire project at 5:41pm. This section is where I'll explain most of the technical details since before this I wasn't really testing the end-to-end process. All I had done before sleeping was checking the dictionary object once a user had right clicked a selection and then clicked "Mine to Anki".

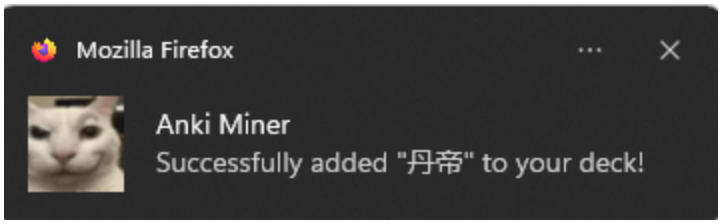
So let's start the technical discussion with this API I keep talking about. While I'm sure there are many different options I decided to go with DeepL for 2 reasons:

1. It was the first result that popped up
2. The registration process wasn't as annoying as Baidu's

With the free tier allowing 500k characters per month I SHOULD be good with these API translations. As for how I'm keeping the API key secure I decided to add a web extension options page where the user can enter the key there. While it is technically being kept as plain text in your browser's storage I think you'd have bigger fish to fry if someone got access to that API key...



The second technical hurdle I want to talk about is how we're going to communicate with Anki since on paper this seems impossible. How on earth would a browser allow a web extension to communicate (and basically modify) a desktop application? It turns out that not only can web extensions send messages to desktop applications there's actually a very streamlined way to do it in Anki. By first installing an Anki addon that listens to a localhost port and enabling CORS to all sources, our web extension can send a message to that localhost destination with our JSON payload and programmatically add a flashcard to our deck!



What a success message looks like

If you're interested in checking out the code (or wanna use it for yourself) then check out the repo here! Otherwise have a great day and I'll see you after this Twice concert!

This blog was finished at 8:47pm - 13 minutes before High Noon